

Revised Generation Interconnection System Impact Study Report

For Queue Project Q22-2205

Warehouse Electric 138 kV
September 20, 2023

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-2205. The Interconnection Customer proposes a 88.0 MW solar photovoltaic facility interconnecting at the Warehouse Electric 138 kV substation in the Pennsylvania area. The request was received on October 11, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 739 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 5 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$2,118,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Lakemont – Weld 138 KV line	Mickleton – Barton 138 KV	116.9	185	5.7
Tunnel – Schuylkill 138 KV	Island Rd – Subestacion Escuela li 138 KV	105.7	259	9.1
Subestacion Escuela li Substation – Island Rd 138 KV	Yreka – Tunnel 138 KV	104.8	257	8.9
Angora – Maplewood 138 KV L	Maplewood Substation – Angora 138 KV Line 2	101.4	290	12.3
Maplewood – Angora 138 KV Cir. 2	Angora – Maplewood 138 KV line	101.4	290	12.3

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Lakemont – Weld 138 KV	Rebuild line with higher-capacity conductor	\$1,948,000	64% (\$1,255,000)
Tunnel – Schuylkill 138 KV	Reconductor line	\$410,000	36% (\$147,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Subestacion Escuela li Substation – Island Rd 138 KV	Reconductor line	\$1,170,000	37% (\$431,000)
Angora – Maplewood 138 KV L	Reconductor line	\$568,000	27% (\$155,000)
Angora – Maplewood 138 KV Line 2	Reconductor line	\$475,000	27% (\$130,000)
	Total allocated to project		\$2,118,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.